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Palaeomagnetism of the Cretaceous El Way and Coloso Formations from the northern Chilean Andes

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SUMMARY

A palaeomagnetic study mainly of limestones of the El Way Formation and fine sandstone of the Coloso Formation near Antofagasta, northern Chile (23.83°S, 70.38°W), gives an early Cretaceous palaeomagnetic pole (76.4°S, 150.7°E, $A_{95} = 6.4°$) for the Chilean Andes. Comparison of this pole with the SAKl pole of Valencio *et al.* (1983) for cratonic South America suggests a relative clockwise rotation of $12° \pm 9°$. Considered in isolation, little confidence can be placed in this possible rotation because of its large error and non-small flattening of $-10° \pm 10°$. However, if the pole is combined with all available late Jurassic and Cretaceous palaeomagnetic poles from the Andes, it is very clear that the poles group according to the geographical location of their sampling areas: Peruvian Andes, hinge part of the Andes and Chilean Andes. The rotation angle between the two blocks calculated from the Peruvian Andes mean pole and the Chilean Andes mean pole amounts to a significant $41° \pm 18°$, assuming the rotation centre is located at the hinge of the coast. A Jurassic sandstone of the Arica Group from Cuya in northernmost Chile seems to have been completely thermally remagnetized by Cretaceous dike swarms which intrude the sandstone. The site mean direction of the sandstone is indistinguishable from that of the dike swarms whose palaeomagnetism has been studied by Nomura *et al.* (1983).

Key words: Andes, Chile, Coloso Formation, Cretaceous, El Way Formation, Palaeomagnetism

INTRODUCTION

The Andes mountain range is a conspicuous feature of South America, and many geophysical and geological studies have been undertaken to investigate the nature and origin of this region. Compilation of palaeomagnetic results obtained so far from the Andes (Heki *et al.* 1983; Kono *et al.* 1985) supports the Arica deflection of Carey (1955), which is an orocline hypothesis for the origin of the deflection in the trend of the Andes mountain belt and the coastline at about 18°S. Other possible large-scale tectonic deformations in the Andes have been highlighted by palaeomagnetic studies in the northernmost (MacDonald & Opdyke 1972; Roperch *et al.* 1987) and southernmost parts (Burns *et al.* 1980; Dalziel *et al.* 1973). In the central Andes around the Arica deflection, it has usually been supposed that the Peruvian block rotated counterclockwise by almost 40° relative to the Chilean block which remained stable. Until recently, Chilean data have been too few in comparison with Peruvian data to conclude unequivocally that there has been no rotation of the Chilean block. However, Beck *et al.* (1986) studied Cretaceous volcanic rocks from the central Chilean Andes and concluded that in addition to counterclockwise rotation of the Peruvian Andes there has also been clockwise rotation of the Chilean Andes. The issue may also have been less clear because we suspect that some of the Chilean Andes palaeomagnetic poles compiled by Heki *et al.* (1983) and Kono *et al.* (1985) have been wrongly classified and may not belong to the Andes, but instead may represent poles for cratonic South America.

The main purpose of this study is to present a new Chilean Andes palaeomagnetic pole of nearly Cretaceous age, and discuss the bearing this has on the rotation problem.

SAMPLING LOCALITIES AND GENERAL GEOLOGY

Sampling localities, a geological map and a stratigraphic section of the Cretaceous El Way and Coloso Formations of northern Chile are shown in Fig. 1. Also shown are the sampling locations of the Jurassic Arica Group sandstone from northernmost Chile and the Jurassic La Negra Formation andesite which outcrops in the vicinity of the El Way and Coloso Formations. The age of El Way and Coloso Formations is early Cretaceous (Hauterivian to Aptian)

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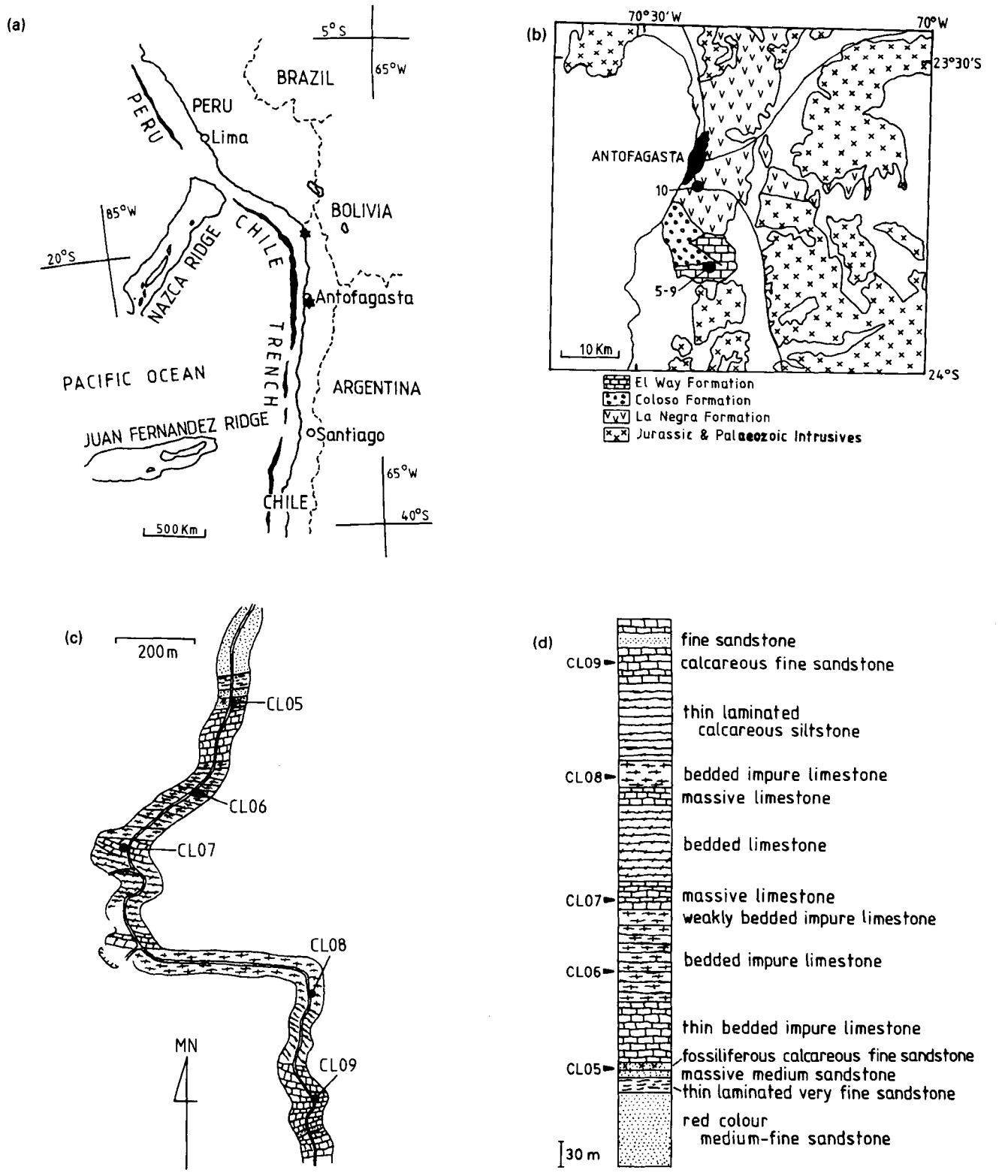


Figure 1. (a) Sampling localities, shown by stars, in the coastal region of the central Andes: Coloso and El Way Formations and La Negra Formation andesite near Antofagasta; Arica Group sandstone near the bending of the Andes. (b) Simplified geology around Antofagasta with palaeomagnetic sites numbered. (c) Detailed map of palaeomagnetic sites 5–9 and (d) stratigraphic columnar section of the Coloso and El Way Formations. The formations strike almost east-west and dip to the south at 20°–30°.

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according to Ferraris & Di Biase (1978). Turner *et al.* (1984), in another geological study of this region, also suggested an early Cretaceous (Neocomian) age for the El Way Formation. They introduced new nomenclature for the formations, but we have used the previously adopted names. The El Way and Coloso Formations in this study are equivalent to their Tobleado and Lombriz Formations, respectively. They also made a palaeomagnetic study of the conglomerate part of the Coloso formation which is lower in the section than the fine sandstone unit which we collected.

We sampled the fine sandstone unit of the uppermost Coloso Formation at one site and limestone to the El Way formation at four sites (Fig. 1c,d). Eight to ten block samples were taken from each site and oriented using a magnetic compass. These sites span a stratigraphic thickness of about 500 m and, as discussed later, we can expect that the five sites are significantly temporally spaced to average out the effects of the palaeosecular variation of the geomagnetic field.

The other two formations sampled are an Upper Jurassic sandstone of the Arica Group near Cuya of northernmost Chile (19.20°S, 70.03°W), which has been thermally affected by intrusive Cretaceous andesite dike swarms, and a Jurassic andesite lava flow of the La Negra Formation near Antofagasta, northern Chile (23.70°S, 70.40°W). The palaeomagnetism of the Cretaceous dike swarms has been reported by Nomura *et al.* (1983).

PALAEOMAGNETIC RESULTS

From each site, six to eight specimens were treated by progressive alternating-field (AF) demagnetization, and at least one specimen was thermally demagnetized. Measurements of the sedimentary specimens were made using a 2G Enterprises cyrogenic magnetometer 740-R at Tokyo Institute of Technology. Most of the andesite specimens were measured with a Schonstedt spinner magnetometer SSM-1A at the University of Tokyo, and the remainder with a ScT cryogenic magnetometer at the Bureau of Mineral Resources, Australia.

Coloso and El Way Formations

An example of a Zijderveld diagram (Zijderveld 1967) for limestone specimens of the El Way formation is shown in Fig. 2 where the direction is tilt corrected. Intensity of natural remanent magnetization (NRM) ranges from 10^{-4} to 10^{-2} A/m and the median destructive field (MDF) is around 20–40 mT. Although thermal demagnetization shows possible acquistion of viscous remanent magnetization (VRM) at temperatures above 500 °C, the results of both thermal and AF demagnetization agree well and define a single component of magnetization.

We could not perform any field tests, such as the fold test, to establish the primary origin of the remanence. Regarding a secondary origin for the remanence, there are two possibilities which we have to consider: VRM induced in the present earth's field (PEF) and remagnetization by heating due to a subsequent geological event such as burial and uplift. Neither are very likely to have occurred within the collected limestones. One of the five sites (CLO5) has an *in situ* site mean direction close to the PEF direction, a feature that is often considered to be diagnostic of a VRM origin for the remanence. However, we do not believe that agreement of an *in situ* direction with the PEF direction is a sufficient condition to reject the site mean direction out of hand. Such a rejection criterion carries the risk that an ancient remanence may be discarded merely because its direction happens, by chance, to agree with the PEF direction before tilting correction. VRM is usually AF demagnetized in fields of 20–30 mT. This is substantiated by the experiment conducted by Irving (1968) who found the difference of the decay curves of AF demagnetization among the oceanic basalts of different polarities to be due to the quick decay of VRM which is parallel to the NRM in a normal polarity sample, while antiparallel in a reversed sample. Nagata (1981) conducted an experiment on the AF stability of VRM of stony meteorites, suggesting coercivities of less than 20–30 mT at most. However, a surprisingly stable VRM with high coercivities of almost 100 mT has been reported for hematite-bearing sedimentary rocks (Biquand

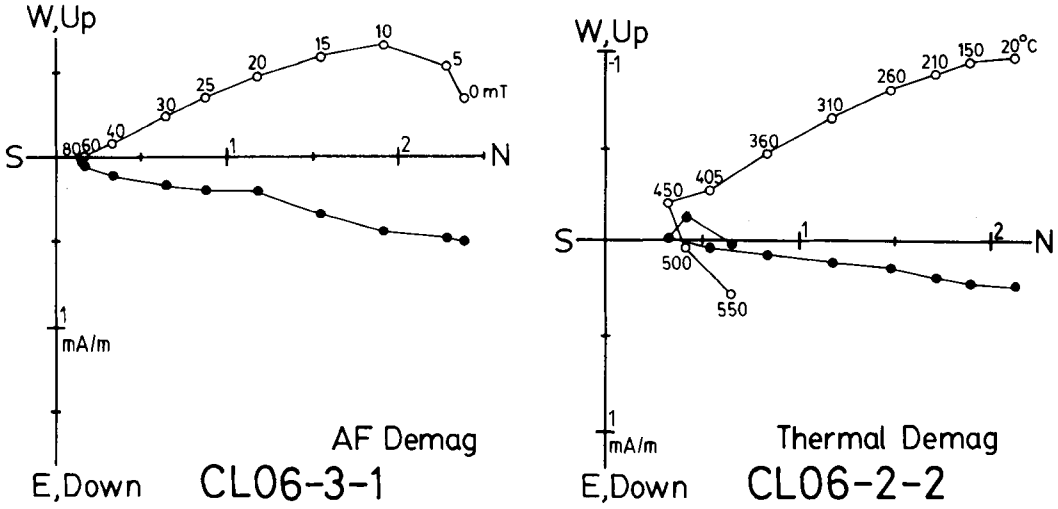


Figure 2. Zijderveld diagrams for limestone specimens of the El Way Formation during progressive AF and thermal demagnetization. The directions are tilt corrected. Open and solid circles show projections onto the vertical and horizontal planes, respectively. Both diagrams show single-component remanences.

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1979) and artificial samples containing fine-grained hematite (Dunlop & Stirling 1977). Kligfield & Channell (1981) reported Mesozoic limestones in the Swiss Alps and showed that they suffered complete remagnetization by VRM during the Brunhes normal epoch. Despite these cautionary observations, we discount the possibility that the remanence of CLO5 is VRM in origin because, in contrast to AF demagnetization, VRM is always unstable to thermal demagnetization, having a low blocking temperature (T_b) of less than 300 °C (Dunlop & Stirling 1977; Kligfield & Channell 1981). The remanences of our samples consistently show a T_b of over 400°C during thermal demagnetization.

The other possibility of a secondary origin for the magnetization is long-term heating associated with burial or metamorphism. This causes build-up of thermoviscous remanent magnetization which can sometimes be very stable depending on the length of time and the temperature of heating (Pullaiah *et al.* 1975). Magnetite-bearing limestones of Devonian age studied by Kent (1985) had two components of remanence: the original component with a higher T_b of up to 500 °C and a secondary component of a lower T_b of less than 300–400 °C. He suggested that the secondary component is thermoviscous in origin and that it was acquired during the late Palaeozoic; he further noted that the maximum T_b of less than 300°–400 °C is always higher than the maximum temperature of about 250 °C to which the limestones had been heated, as indicated by the alteration index. Taking this into account and the fact that thin sections of the El Way limestone show no indication of alteration, we conclude that the remanence of the limestone is not thermoviscous in origin. Long-term heating at fairly high temperature would be necessary to induce such a stable thermoviscous remanence and destroy the original one.

The site mean directions from the El Way and Coloso Formations are shown in Fig. 3 and summarized in Table 1. The single polarity of the directions introduces some uncertainty as to whether palaeosecular variation (PSV) of the geomagnetic field has been averaged out. The reversal test (McElhinny 1973) is often used to establish this because the phenomenon of reversal indicates that the sample collection does span a sufficient length of time. However, we cannot apply it in this instance. Nevertheless, we think the effect of PSV is sufficiently averaged out because the five sites cover an almost 500 m thickness of limestone. Using a typical sedimentation rate of 1–20 m/Myr (Lowrie & Heller 1982) for this type of material, the section must represent a time span of the order of millions of years, even allowing for the fact that the limestone of the El Way Formation might have a somewhat higher sedimentation rate because it contains minor interbeds of siltstone and fine sandstone.

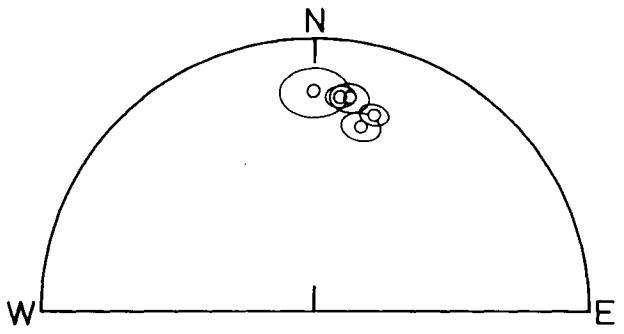


Figure 3. Lambert equal-area projection of site mean directions of the Coloso and El Way Formations. The directions are all normal with negative inclinations and are denoted by small open circles. Their associated circles of α_{95} are also shown.

Table 1. Palaeomagnetic site directions.

		Field direction			Virtual geomagnetic pole			
Site	*N*	*I*	*D*	α_{95}	Lat. °S	Long. °E	*dp*	*dm*
Coloso and El Way Formations (23.83°S, 70.38°W)								
CL05	8	−21.3	359.9	8.6	77.2	109.3	4.7	9.0
CL06	10	−22.8	6.8	4.0	76.4	139.1	2.2	4.2
CL07	9	−31.7	13.8	5.2	75.5	174.8	3.3	5.8
CL08	8	−26.8	16.5	3.8	71.7	170.9	2.3	4.1
CL09	10	−22.9	9.1	5.0	75.3	147.0	2.8	5.3
Mean		−25.2	9.1	6.8	76.4	150.7	$A_{95} = 6.4$	
Arica Group sandstone (19.20°S, 70.03°W)								
CL02	8	17.1	166.7	14.7	73.4	57.3	7.9	15.2
La Negra Formation andesite (23.70°S, 70.40°W)								
CL10	4	−27.9	9.9	5.7	77.1	157.7	3.4	6.2

Key: *N*: number of samples; α_{95} and A_{95}: 95 per cent confidence circles of direction of magnetization and palaeomagnetic pole, respectively; *dp* and *dm*: error parameters of oval of 95 per cent confidence of VGP.

Another method often adopted to ascertain the removal of PSV is to check the magnitude of the angular standard deviation (ASD) of the observed virtual geomagnetic poles (VGP) (Cox 1970). If the calculated ASD is too small, there is a possibility that the sampling sites represent too short a period of time. The ASD calculated from the five VGPs of the El Way and Coloso Formations is 6.8° with upper and lower 95 per cent confidence limits (Cox 1969) of 11.1° and 4.8°, respectively. This ASD value is much smaller for the calculated palaeolatitude of about 15°S than that reported over the past 5 Myr (McElhinny & Merrill 1975). However, we do not think this value is unreasonable for two reasons: (1) the ASD represents the PSV due to dipole wobble only, because the slow sedimentation rate of the limestone would average out the contribution due to variation in the non-dipole components; (2) the PSV of the dipole wobble in the Cretaceous is expected to be small according to McFadden & McElhinny (1984) who fitted their new PSV model F to the data for 45–110 Ma and found that the ASD of dipole wobble must necessarily be zero to obtain a fit that is physically realistic.

The overall mean direction for the four sites of the El Way Formation and one site of the uppermost Coloso Formation is a declination (*D*) of 9.1°, inclination (*I*) of −25.2° and a circle of confidence of 95 percent, α_{95} or A_{95} (Fisher 1953), of 6.8° which gives an early Cretaceous palaeomagnetic pole of 76.4°S, 150.7°E, $A_{95} = 6.4°$ for the northern Chilean Andes.

La Negra Formation

We found a complex remanence comprising two antipodal directions in individual specimens of an andesitic lava flow of the Jurassic La Negra Formation. A Zijderveld diagram of a typical thermal demagnetization is shown in Fig. 4; the directions have been tilt corrected according to the bedding

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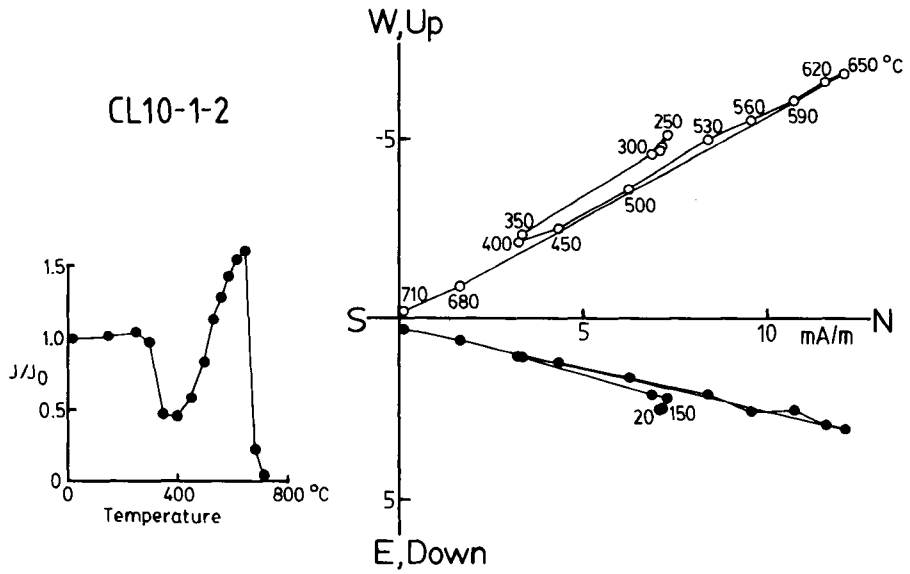


Figure 4. Zijderveld diagram for a specimen of andesite from the La Negra Formation during progressive thermal demagnetization in air. The notation convention is the same as in Fig. 2; the data are tilt corrected. The inset shows a normalized decay curve of remanence intensity during demagnetization.

of the underlying sandstone. This peculiar behaviour of the remanence is similar to, but much clearer than, that reported by Turner *et al.* (1984) for the conglomerate part of the Coloso Formation. They interpreted the reversed component to be original and the normal to be an overprint. We think, however, that rock magnetic phenomena during thermal demagnetization, including the possibility of self-reversal, is a more likely explanation in our case because the directions within specimens are completely antipodal and their initial susceptibilities change by a large amount with increasing temperature during thermal demagnetization. In any case, the La Negra Formation and the conglomerate part of the Coloso Formation, both of which are lower in the stratigraphic sequence than the El Way Formation, must have suffered alteration as a result of reheating associated with a later localized mineralization effect (Turner *et al.* 1984). Thermal demagnetization of four samples of the lava flow gives a site mean direction of $D = 9.9°$, $I = -27.9°$, $\alpha_{95} = 5.7°$. Although this is close to the result of Turner *et al.* (1984)—$D = 21°$, $I = -36°$, $\alpha_{95} = 6.8°$—further discussion is difficult because no mention is made of the tilt correction applied to their data.

Arica Group

We report on a Jurassic sandstone of the Arica Group sampled near Cuya, northernmost Chile (Fig. 1a), because it is interesting to compare the remanence direction with the palaeomagnetic result obtained by Nomura *et al.* (1983) on the Cretaceous Cuya dike swarms which intrude the sandstone. We had expected to obtain a Jurassic result from the host rock and collected samples of the sandstone 5–30 m away from the dikes which intrude it every 30–50 m. However, the site mean direction of the sandstone ($D = 166.7°$, $I = 17.1°$, $\alpha_{95} = 14.7°$) is indistinguishable from the mean direction ($D = 166.5°$, $I = 17.4°$, $\alpha_{95} = 10.4°$) obtained by Nomura *et al.* (1983) based on 25 dikes. This strongly suggests that the sandstone has suffered complete thermal remagnetization by the dikes at the time of their intrusion. The sudden decrease of remanence around 300–350 °C during thermal demagnetization, compared to the linear decay at high coercivities during AF demagnetization (Fig. 5), supports this interpretation. The sandstone results established the remanence of the dikes as primary.

DISCUSSION

The early Cretaceous pole obtained in this study for the northern Chilean Andes (76.4°S, 150.7°E, $A_{95} = 6.4°$), when compared with the SAKI (85°S, 75°E) reference pole of Valencio *et al.* (1983) for cratonic South America, suggests a relative clockwise rotation of $12° \pm 9°$ and a flattening of $-10° \pm 10°$. These parameters are defined respectively as $D_o - D_c$ and $I_e - I_o$, where D_o, I_o are the observed direction, and D_e, I_e are the expected direction from the cratonic reference pole (Beck 1980), and the errors are the 95 per cent confidence limits. There is little purpose in discussing this possible clockwise rotation of the northern Chilean Andes solely on the basis of our result because of the large errors in the rotation and flattening values. Even if the errors were small, it would still be difficult to conclude from a single result that the origin of the rotation is due to large-scale crustal block movement rather than some local tectonic effect. Palmer *et al.* (1980b) studied the 157 Ma Camaraca Formation near the hinge region and obtained 28° and 44° counterclockwise rotations when they compared its pole, respectively, with a cratonic South American reference pole and an expected reference pole obtained by rotating African data to South America so as to close the South Atlantic. They concluded that it is difficult to discriminate between whether the effect is due to a local or larger-scale tectonic cause. Irwin *et al.* (1987) studied Jurassic mafic dikes from central Chile and concluded from the palaeomagnetic results that there has been no tectonic

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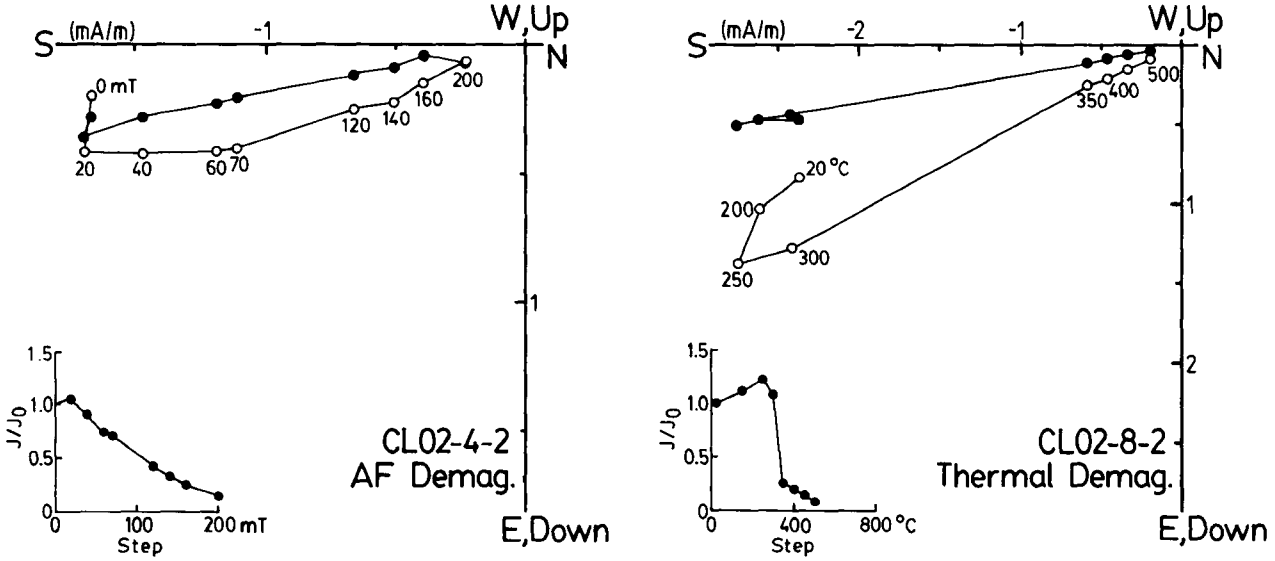


Figure 5. Zijderveld diagrams of progressive AF and thermal demagnetization of sandstone specimens from the Arica Group. The data are tilt corrected. The sudden decrease of intensity around 300–350°C during thermal demagnetization indicates that the sandstone has suffered from heating during dike intrusion. The remanence direction is exactly the same as that obtained by Nomura *et al.* (1983) in a palaeomagnetic study of the dike swarms.

rotation. However, on the basis of geological arguments, they inferred an allochthonous origin for a metamorphic complex which the dikes intrude. They further suggested that the clockwise-rotated declination of magnetization found by Beck *et al.* (1986) for the ash-flow tuffs of the Chilean Andes is likely to have been caused by local crustal extension related to magmatism. Recently, Jesinkey *et al.* (1987) reported a late Palaeozoic palaeomagnetic pole, from the central part of the Chilean Andes, which lies on the apparent polar wander path of cratonic South America defined by Irving & Irving (1982). They claimed that the Andes had therefore been part of the craton since the

Table 2. Summary of Cretaceous and Jurassic palaeomagnetic poles for central Andes.

		Locality			Pole position					
Formation	Age	°S	°W	Site	°S	°E	A_{95}	*dp*	*dm*	Ref.
Peruvian Andes										
Coastal volcanics	K	10	77	12	68.0	359.5	5.3			4
Herradura, Vinchos, Moracoche area	K	11	76	4	63	30	8			1
Puente Piedra Formation	Kl-u(90 Ma)	11.9	77.1	12	73.3	359.3		2.0	3.7	8
Yumagual Formation	Kl-u(Cenomanian)	5.9	78.2	1	66.7	339.7		3.2	5.7	4
Chulec Formation	Kl-u(Albian)	7.1	78.3	1	52.0	1.6		4.7	8.8	4
Pariatambo Formation	Kl-u(Albian)	7.1	78.3	1	35.9	0.8		3.1	5.8	4
Hinge Region										
Arica dike swarm	K	18.6	70.3	19	77.2	352.4	3.3			7
Cuya dike swarm	K	19.2	70.0	25	74.1	49.0	7.8			5
Atajana Formation	Kl	18.6	70.3	1	74.7	37.9		2.4	4.4	7
Camaraca Formation	Ju(157 Ma)	18.6	70.3	33	70.5	9.9	5.9			3
Chilean Andes										
La Serena area	K (Hauterivian–Coniacian)	29.8	70.9	30	81	209	4.5			2
San Fernando tuff	Kl-m(100 Ma)	34.6	71.1	22	76.3	192.5	7.1			9
El Way and Coloso Formations	Kl(Neocomian)	23.8	70.4	5	76.4	150.7	6.4			*
Coloso Formation	Kl	23.7	70.4	13	70	192		5	8	6
Coastal intrusives	Ju(145 Ma)	33	71.5	10	80.3	207.8	9.5			10
Coastal intrusives	Jm(170 Ma)	33	71.5	8	85.9	36.7	11.9			10

Key: K(l, u, or l-u), Cretaceous (Lower, Upper, or upper part of Lower Cretaceous to lower part of Upper Cretaceous); J(m or u), Jurassic (Middle or Upper).
References: *, This study; 1: Creer (1970); 2: Palmer *et al.* (1980a); 3: Plamer *et al.* (1980b); 4: Heki *et al.* (1983); 5: Nomura *et al.* (1983); 6: Turner *et al.* (1984); 7: Heki *et al.* (1985a); 8: May & Butler (1985); 9: Beck *et al.* (1986); 10: Irwin *et al.* (1987).

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Carboniferous, and suggested that the discordant directions reported from Mesozoic and Cenozoic formations of the Andes might be due to either localized tectonic movement or artificial errors in palaeomagnetism because most data lack a field test of their stability of remanence.

The discussion is further complicated by the very streaked distribution of the South American cartonic reference poles for the Jurassic and Cretaceous as pointed out by Beck *et al.* (1986). This behaviour of the pole has been interpreted by Valencio *et al.* (1983) in terms of rapid polar movement and hairpins.

If, however, all the data are examined collectively, the difference in pole positions between the Peruvian and Chilean Andes becomes conspicuous. The combined data are summarized in Table 2 and plotted in Fig. 6. Cretaceous and Jurassic south poles are shown by three different symbols: circles, squares and triangles correspond to data derived from the Peruvian Andes, hinge region and Chilean Andes, respectively. Diamond-shaped symbols with hatched A_{95} circles are reference poles from the stable craton of South America defined by Valencio *et al.* (1983): 1, SAKel (late Jurassic to early early Cretaceous or 157–121 Ma); 2, SAKl (early Cretaceous or 142–101 Ma); 3, SAKl-u (late early to early late Cretaceous or 114–75 Ma). Open symbols along the coast show sampling site locations. The mean position of Peruvian Andes poles is 61°S, 2°E with A_{95} of 13.1°. Chilean Andes poles fall about 40° from these at 80°S, 187°E with A_{95} of 7.4°, and those from the hinge region fall in between at 75°S, 23°E with A_{95} of 8.2°. This clear grouping of Andes poles according to geographical location of their sampling sites is surely not due to rapid movement of the pole with respect to South America as suggested by Valencio *et al.* (1983), because there is no relationship between the pole groupings and their ages. The difference between the mean positions of the Peruvian and Chilean Andes poles strongly suggests that relative rotation has occurred between the two regions of the Andes as argued previously. If the pole of rotation is assumed to be the hinge region (18°S, 290°E), the angle of relative rotation amounts to a significant $41° \pm 18°$. This almost agrees with the deflection angle of the coastline at the hinge.

We wish to emphasize that it is by no means obvious that only the Peruvian Andes have rotated as suggested previously (Heki *et al.* 1983; Kono *et al.* 1985). As already pointed out by Beck *et al.* (1986), it is more natural to consider that the Chilean Andes have also rotated, but in a clockwise sense as clearly indicated in Fig. 6. The reason for the suggestion that only the Peruvian Andes have rotated might be due to the fewer poles available for the Chilean Andes to date and possible mistaken classification of some poles as Chilean Andes poles by Heki *et al.* (1983), which might be better classified as craton poles.

In view of the above discussion, we think it very unlikely that most of the deflected palaeomagnetic directions from the Andes are caused by either local tectonic movement or artificial error, as Jesinkey *et al.* (1987) concluded. On the contrary, palaeomagnetic results suggest a simple bending of the Andes mountain belt such as that shown in cartoon form in Fig. 7. Regarding the timing of the bending, Heki *et al.* (1985b) suggested possible continuation of rotation until late Tertiary, based on their result from a palaeomagnetic study of the Neogene Ocros dike swarm of southern Peru which still shows a slightly deflected direction of magnetization. However, Tsunakawa *et al.* (1987) showed that rotation

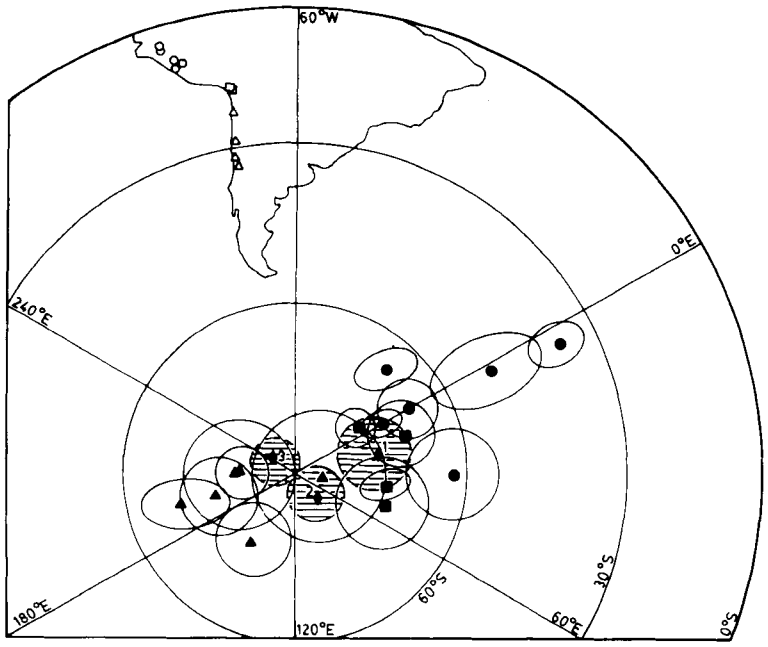


Figure 6. Lambert equal-area projection of palaeomagnetic south poles of Cretaceous and Jurassic rocks from the central Andes which are summarized in Table 2. Three different symbols are used according to the three groups of sampling site locations: circles, squares and triangles correspond to data from the Peruvian Andes, hinge region of the Andes and the Chilean Andes, respectively. Open symbols along the coast of the Andes show site locations and closed ones with A_{95} circles or ellipses show palaeomagnetic south poles. Diamond-shaped marks with hatched A_{95} circles are reference poles from cratons in South America defined by Valencio *et al.* (1983): 1, SAKel (late Jurassic to early early Cretaceous or 157–121 Ma); 2, SAKl (early Cretaceous or 142–101 Ma); 3, SAKl-u (late early to early late Cretaceous or 114–75 Ma).

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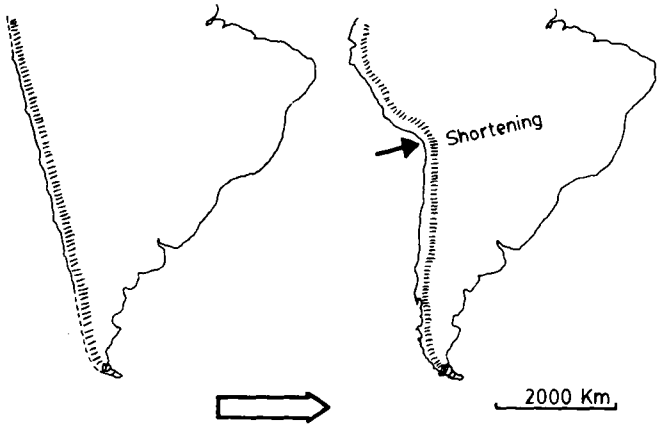


Figure 7. Bending of the Andes mountain belt. Block rotation of both the Peruvian and Chilean Andes is the most probable deformation style that occurred in the central Andes.

must have ceased prior to the Late Miocene on the basis that there is no significant difference between four Tertiary poles when compared: their new pole, that of Heki *et al.* (1985b), and two cratonic poles of Creer & Valencio (1970) and Vilas (1981).

In spite of the clear evidence from palaeomagnetism, the bending of the Andes mountain belt might be a difficult aspect to incorporate into models of its formation. Three main kinds of models have been proposed, as represented by the following three authors. James (1973) proposed that subducting oceanic plate induced magma generation at its upper surface and that the centre of magma activity migrated eastward with time, forming the thicker crust now found in the east. Uyeda & Kanamori (1979) classified subduction zones of the world into two types: Chilean type and Mariana type. Formation of the Andes is a typical case of the Chilean type in which a young, light, oceanic plate subducts at a shallow angle and strongly interacts with the continental crust, causing folding and thickening of the crust. Suarez *et al.* (1983) proposed that reverse faulting has occurred repeatedly at the eastern margin of the Andes, causing shortening and thickening of the crust. However, none of these models includes the constraint of bending identified by palaeomagnetism. Recently, Kono *et al.* (1988) proposed a new model based on many geological and geophysical data, including their new gravity map. In their model, frequent magma intrusion along the western belt of the Andes, due to the low angle of subduction, is the cause of thick crust, and along the eastern belt, crustal shortening has occurred due to dominant horizontal compression. However, again a block rotation mechanism of the Andes has not been considered. Recent photo-geologic work on Landsat images and Shuttle Imaging Radar by Allenby (1987) would seem to be the only study which refers to block rotation of Peru. According to Allenby, the greater *resistance of the Brazilian shield* to eastward movement of the coastal region, than the southern part of the continent, caused a sinistral fault zone to develop over the Arica hinge region and extend into the inner shield zone, and also resulted in counterclockwise rotation of the northern part of the Andes.

It is beyond the scope of this paper to give a detailed discussion of the formation history of the Andes. Our ideas concerning the simple bending of the Andes, as shown in Fig. 7, will be discussed elsewhere (Tsunakawa in preparation). Suffice it to say that bending of the Andes mountain belt is easily explained if the spherical shell tectonics concept of Yamaoka *et al.* (1986) is taken into account. We would, however, like to point out here that, according to Amano *et al.* (1988), collision of the Nazca ridge with South America could not contribute to the bending of the Andes. They showed that the collision of such aseismic ridges is only effective in uplift of the local coast region without any large tectonic deformation.

CONCLUSION

A new palaeomagnetic pole for the northern Chilean Andes of early Cretaceous age is reported, which shows a possible clockwise rotation of $12° \pm 9°$ when compared with cratonic South America reference poles. When combined with all data from the central Andes, poles group clearly according to their geographical origin: Peruvian Andes, Chilean Andes and hinge region. It is clear that this grouping of Andes poles is not due to grouping by age, which might occur from the rapid polar wander or hairpins suggested by Valencio *et al.* (1983). If the pole of rotation is assumed to be the hinge region, the bending angle amounts to $41° \pm 18°$, which is close to the deflection angle of the coastline at the hinge. Despite the difficulty in defining the exact position of the cratonic reference pole because of the streaked distribution, it seems most probable that both the Peruvian and Chilean Andes have rotated as pointed out by Beck *et al.* (1986).

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